Implications of IOT

The Internet of Things enables physical assets to become elements of an information system, creating the ability to capture, compute, communicate, and collaborate in novel ways. The Internet of Things is still in its infancy, but its impact is growing quickly. It is predicted that there will be between 25 billion and 50 billion connected devices by 2025. Governments, policy makers, and businesses have the chance to accelerate the enormous opportunities makers, and the companies that supply IoT technology, beginning with implications for users of IoT associated with the Internet of Things, even as they work to address the risks that accompany the technology. We now look at the implications of IoT for different stakeholders—companies and consumers that use IoT technology, policy technology and policy makers followed by implications for the companies that will supply IoT technologies.

Implications for users and policy makers

As discussed earlier, the introduction of IoT systems into the homes, cities, factories, and other settings has the potential to create value in new ways and to alter how people live and work. IoT applications will affect the full range of stakeholders, from individuals, whose home and work routines will change, to corporations that will have new ways to compete (as well as new competitors). The Internet of Things can enable new business models, including those that monetize the data generated by IoT technology.

- **Consumers**

The Internet of Things offers substantial *benefits* for consumers as well as a new set of *risks*. On the benefit side, IoT applications offer the potential to *drive down the costs* and *enhance the quality of goods and services*, as manufacturers, retailers, and other suppliers employ IoT systems to *improve product design and operations*. Another source of value for consumers will be *greater consumer convenience and time savings*. As they travel, consumers can benefit from IoT-managed roadways, *self-driving cars, real-time public transit information, and planes that take off and land on schedule*. In stores, shoppers could experience a new level of service, savings via *targeted in-store offers, and automated checkout*. At home, they can offload *housework* to smart appliances, *save money on energy* use, and *protect* themselves with IoT-enabled *security systems*. And, most importantly, lives can be extended and improved by IoT *health and wellness applications*.

Given the value associated with interoperability, consumers can choose to purchase systems that are interoperable so that they can maximize benefits. In some cases, this might mean purchasing different systems from one vendor; in others, it might mean purchasing systems from multiple vendors that use common standards.

IoT benefits also come with a new range of risks. The Internet has already raised concerns about consumer privacy, and the mobile Internet (Smartphones) has heightened those concerns by collecting passive tracking data. IoT devices will go beyond that. Wellness monitors can not only report where a person is but also can gather data about what the person is doing. Health IoT systems will collect information about the consumer's weight, eating habits, exercise

regimes, drugs used, and diseases. In the workplace, IoT systems can be used to monitor and manage employee performance, including by "reading" facial expressions to assess a worker's emotional state.

Consumers will need to be diligent about monitoring the data that are being gathered about them. When they sign up for services, they should bear in mind what kind of permissions they are granting for the use of their data, as well as what kind of access they allow to the data they are generating. Consumers can also participate in policy-making dialogues as regulators determine how detailed personal information can be used and how people can exert control over their personal data.

Finally, consumers will have to manage information overload. The physical and psychological effects of constant Internet and Smartphone use are beginning to be acknowledged, as are "context switching costs" when the brain is interrupted by an email alert or an incoming tweet. Consumer IoT systems can be yet another source of overwhelming data and distraction. Consumers will need to think about which IoT systems to adopt and how to integrate the use of those systems into their lives. Many may avoid systems that provide a constant stream of information and instead opt for those that present information only when it is relevant.

- **Employees**

As with other productivity-improving technologies, IoT will affect workers in different ways. Some types of workers are likely to benefit as the Internet of Things creates new needs for specific skills. Workers who have skills in developing and deploying IoT systems will find themselves in greater demand. During the period of rapid growth in IoT deployments, developers and data scientists are likely to benefit the most. Workers with systems integration skills also will be in demand, because of the high degree of customization required for many IoT implementations.

Over time, applications enabled by IoT are likely to enable the automation of an increasing number of tasks currently performed by service workers. For example, IoT could make it possible to economically automate some food preparation. Self-guided cleaning robots could reduce the demand for office and home cleaners, and IoT-based security systems could reduce the number of security guards needed to patrol commercial spaces. Automated checkout sharply reduces the need for cashiers. In general, manual work will come under increasing pressure from IoT and smart machines, but IoT will open up some new employment opportunities, too. Workers who can install and maintain sensor networks, autonomous vehicles, and other IoT systems will be needed.

- **Organizations that use the Internet of things**

The businesses and other organizations that use IoT systems are faced with a multitude of opportunities—and maybe just as many challenges. Depending on the industry, IoT applications can have far-reaching effects on how companies operate and may even force changes in business models. Some businesses (automobile manufacturers, for example) need to master both use of IoT in their operations and in their products. We look at implications for businesses in three ways:

- using IoT to improve current business models,
- new business models enabled by IoT and
- Organizational issues related to IoT implementation

i) Using IoT to improve current business models

First, companies need to identify and prioritize opportunities for IoT to enable them to improve performance in their current operations and lines of business. These opportunities are extensions of the opportunities that have been identified with the use of big data and advanced analytics. IoT data will allow companies to personalize services based on consumer behavior, usage, and context. For example, a clothing store that recognizes individual shoppers and knows their purchase histories can suggest outfits that align with their tastes when IoT sensors detect that they have entered the store, saving the consumer from the hassle of browsing through the store's entire inventory. However, even in consumer-related industries, the majority of potential benefits will be in B2B applications, such as manufacturing and supply chain.

It is also important to keep in mind the substantial potential benefits of deploying IoT in businesses in developing markets as well as in advanced economies. Depending on conditions in the specific developing economy, IoT applications can be used to create large amounts of value, often from different sources than those in advanced economies.

Interoperability is key to obtaining much of the value from the Internet of Things. As sophisticated and powerful customers, companies can demand interoperability when they write specifications and procure IoT systems.

As companies identify and prioritize opportunities to use IoT applications, they should inventory all potential sources of data, particularly those they might already own but have not yet fully exploited. IoT is an incredible source of data generated in the course of operations that can be used for other purposes. As noted, in IoT applications, most data, including so-called exhaust data (data that is generated as a "byproduct" of IoT instrumentation), is not yet used or is used only for anomaly detection and/or real-time control. Far more value remains to be captured in analyzing the data for optimization and prediction.

Following are eleven broad categories of applications through which IoT can improve performance and create value for business users across settings and sectors.

The first four categories of applications fall under operations optimization.

- *Inventory management*. Tracking inventory and supplies in retail environments, factories, warehouses, and hospitals
- *Condition-based maintenance*. Deploying sensor data to determine when maintenance is actually needed, reducing breakdowns and costs
- *Human productivity*. Using IoT to teach skills, redesign work, and manage performance

- *Other operations optimization opportunities*. Remotely monitoring and tracking equipment, as well as automatically adjusting machinery based on IoT data, among other uses
- *Sales enablement.* Using IoT usage data to generate new sales
- *Health management*. Improving health and wellness using IoT monitoring data
- *Energy management*. Using IoT sensors and smart meters to better manage energy
- *Environmental management.* Improving stewardship of the environment using IoT technology, such as using sensor data to reduce air pollution
- *Safety and security*. Using IoT sensors to mitigate safety and security risks
- *Product development.* Employing IoT usage data forresearch and development
- *Autonomous vehicles*. Adopting fully or partially self-driving cars, trucks, and public transportation vehicles.

The largest source of potential impact (60 percent) is operations optimization, such as inventory management and condition-based maintenance, which requires optimization and prediction. But each company will have to analyze and prioritize its own opportunities.

ii) New business models for companies that use IoT

In addition to gaining operating efficiencies and the other benefits that companies can derive from using IoT in their existing businesses, companies can create new business models based on IoT e.g.:

- *New pricing models*. IoT creates a continuous, fine-grained stream of real-time data. These data enable both microsegmentation and insights into consumer willingness to pay. Customized, dynamic pricing models based on specific customer preferences, as well as the usage context, can provide benefits for both companies and consumers. For example,
    - auto insurance companies have begun to sell policies with premiums based on sensor data about actual driving behavior, rather than the demographics of drivers.
    - A company selling smart lighting fixtures could charge customers based on the amount of money they save.

- *Service-based business models.* The Internet has enabled the "as a service" business model for IT infrastructure and software. The Internet of Things enables "anything as a service" business models for all kinds of other products, potentially letting many kinds of companies shift from selling products to selling services based on those products. This model can transform large capital expenditures into a pay-by-usage operating expense. Examples of this trend are proliferating. They include selling:
    - "power by the hour" rather than gas turbines,
    - transportation services rather than automobiles,
    - pages printed rather than laser printers.

IoT technology not only enables the providers of these services to charge by usage, but it also enables these companies to better maintain and upgrade the equipment that is used in these services, removing the maintenance burden from customers and creating the basis of long-term relationships. IoT-based service businesses can also use consolidated customer usage data and data about individual customers for cross-selling, guiding product development, and other purposes. Furthermore, IoT can allow products to become better while in service (the opposite of depreciating in value), a concept Hal Varian has dubbed "product kaizen".

- *Monetization of IoT data*. The exhaust data generated by IoT applications can become a profit center itself. For example, almost any data describing consumer behavior can be of great value to marketers. Data about physical assets such as buildings and vehicles can be used to assess insurance risks. One company's data exhaust could be another company's gold mine, and that value could be monetized by the originator of the data. Of course, privacy, confidentiality, and ownership rights over such data (for instance, consumer purchasing data) are issues that would have to be addressed to create such business models.

iii) Organizational implications of IoT

Companies building strategies around the Internet of Things will need to develop new organizational capabilities and, in many cases, adopt different behaviors. For companies to make the most of IoT technology, their organizations must embrace data-driven decision making and capabilities and create environments in which relevant information is made available to the right decision makers (which could be people or algorithms) in a timely manner. This may require training workers in new skills and investing in new software systems to capture and analyze IoT data. Indeed, all of the implications of big data and analytics apply to IoT data.

Whatever the application, the value of IoT data depends on effective use. Organizations will need to become adept at data-driven decision making, which requires analytical rigor. As companies build up their data analytics capabilities (both technologies and talent) to handle the flood of IoT data, the analytics experts must be connected with decision makers as well as frontline managers; both groups need to learn how to apply insights from large, real-time data streams in a timely manner. And when real-time action needs to be taken on a large scale—such as optimizing machine control across an entire factory or providing real-time marketing offers to shoppers—organizations will have to develop processes that enable algorithms to make decisions automatically, while allowing decision makers to monitor metrics and set policy.

The role of the IT function within organizations will also be transformed by the Internet of Things. The scope of responsibilities of IT departments will no longer be primarily in data centers and managing desktop and mobile devices. IoT applications will extend IT systems into physical assets, facilities, and inventory. Thus, more than ever, the IT organization will have to be closely aligned and even integrated with the operating parts of the business. And the metrics of success for IT will have to measure more than technology, extending to the performance of the business itself.

Synthesizing data from different sources will take more than interoperable systems; it will also require collaboration among leaders and teams in different parts of the organization. Owners of various IT systems (chief financial officers and chief marketing officers, for example) must be receptive to linking up their systems and providing data access across the organization.

Many IoT applications will also require collaboration. For example, in health care, IoT data will need to be shared among doctors, hospitals, insurance companies, device manufacturers, and software providers. In some instances, IoT applications may even depend on competitors working together. In mining, for example, multiple small operators would need to share data in order to optimize their operations in the same way as a large operator. To get insights from data on the characteristics of an ore body in one mine and make effective predictions requires a large comparative data set, which small operators may not be able to get unless they pool their data.

- **Policy makers**

For IoT applications to achieve their full potential, issues in several areas must be resolved: privacy and confidentiality, intellectual property, security, and interoperability. In each of these areas, government action can be an enabler. The explosion of IoT-based data about what companies and consumers are doing raises important concerns about personal privacy and corporate confidentiality. Who has access to and use of data will become a major issue since many forms of data collection—license plate scanners to catch speeders, for example—do not require consent. Governments can help to establish practices and constraints about data collection, access, and usage, especially for data generated in public spaces. A closely related issue is data rights of various actors in society. What rights do consumers have over data that is generated by or about them? These are also questions in which government has a role to play, through legislation, rule making, policy, and judicial rulings.

Leadership by policy makers can also help address security issues. Governments can invest in research to address security risks, convene and fund multi-stakeholder centers to set standards and share information, model good security practices, and craft thoughtful rules to encourage security management and punish bad actors. Finally, governments and other policy-making organizations, such as industry groups, can encourage and support the development of standards that will enable interoperability of IoT devices and systems.

Implications for technology suppliers

The Internet of Things represents a major opportunity for incumbent technology suppliers as well as for emerging players. The market for IoT components and systems grew 160 percent in 2013 and 2014, and growth is projected to remain above 30 percent a year through 2025. As in other technology markets, the IoT market will have a variety of players and strategies. Some suppliers will compete by offering distinctive technology; others will offer distinctive data. There also will be companies that establish technology platforms and those that specialize in offering comprehensive, end-to-end solutions. The opportunities to assume these roles vary by type of player. We also look at the different sources of competitive advantage for technology suppliers, how we expect value to be distributed among

different types of technology suppliers, and how technology suppliers can play across multiple vertical markets—or find it difficult to do so.

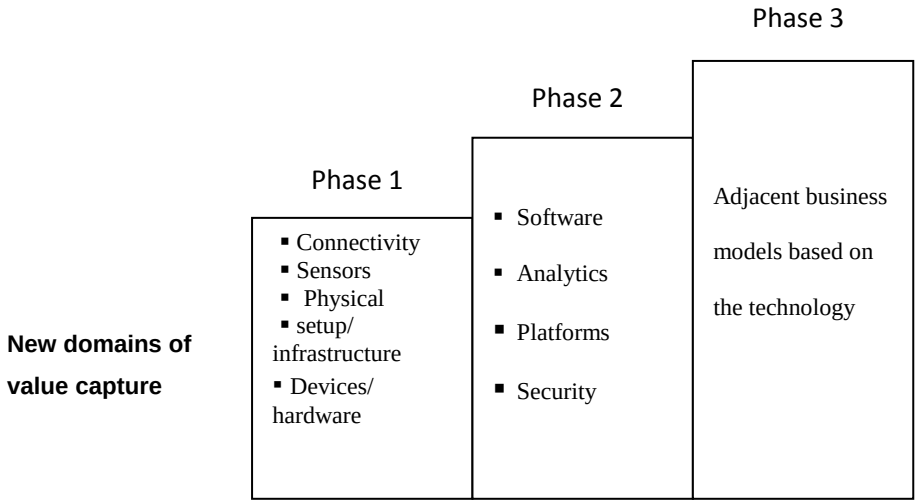
- **The shifting IoT industry value chain**

We expect that, as in other Internet applications, users (businesses and consumers) will capture most of the value created by IoT use—potentially as much as 90 percent in 2025. But given the huge amount of potential value that IoT can generate, this still leaves a substantial market for the suppliers of IoT hardware, software, and services. As the IoT industry evolves over the next ten years, we expect the division of value among players to shift, with an increasing share going to suppliers of software and analytics. Suppliers of foundational technologies such as hardware and IoT device clouds as well as installers of IoT systems are likely to capture less value in 2025 than they do today. Makers of packaged software and applications developers are likely to gain. In 2025, we would expect software and service providers to capture up to 85 percent of the IoT revenue generated by suppliers.

- **Phases in evolution of IoT technology**

The shift in value among IoT suppliers could follow a course similar to the pattern of previous technology waves. In the personal computer industry and in the history of the Internet itself, there were three overlapping periods in which value shifted toward three types of companies. In the first phase, suppliers of infrastructure and hardware dominate. During the Internet era, companies such as Cisco and 3Com provided the basic building blocks of infrastructure. In the second phase, some core services, built on the infrastructure, grow quickly to global scale. In the case of the Internet, companies such as Google and Yahoo provided services such as search. In the third phase, adjacent business models appear. On the Internet, this has included companies such as Amazon, which pioneered Internet retailing, and Airbnb, in peer-to-peer lodging services. Today, the Internet of Things is somewhere between the first and second phases. Some companies are focusing on the basic building blocks, such as connectivity, sensors, and other IoT devices, while others are beginning to specialize in software, analytics, and security that could begin to scale

IoT supplier industry could evolve in three phases



- **Bases of competition for IoT technology suppliers**

For companies to establish a market presence and maintain share over the long term, they will need to understand how their strategies and strengths fit within the three phases of the industry evolution. Across these three phases are at least four distinct bases of competition that can lead to disproportionately large value capture opportunities for individual companies: distinctive technology, distinctive data, technology platforms, and the ability to provide complete solutions.

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- *Distinctive technology*. For broad adoption of IoT technology, several basic technologies are needed, such as low-cost, low-power sensors and low-cost connectivity. Companies that have proprietary intellectual property in these areas can build strong, sustainable positions. In the same way that ARM was successful in the smartphone era because of its distinctive low-power chipsets, players that develop distinctive IoT-relevant technology will be at an advantage.
- *Distinctive data*. There is significant promise for the owners of valuable proprietary IoT data. For instance, a manufacturer of industrial equipment can collect hundreds of terabytes of streaming data daily from sensors used by its customers, enabling the manufacturer to identify anomalies before they cause equipment shutdowns and power outages, and also enabling the manufacturer to provide predictive maintenance that others with less access to data would be challenged to provide.
- *Platform providers*. In other IT markets, companies have created broad-based software-enabled platforms upon which third parties build applications. Platforms are also a means to provide interoperability among applications. A successful platform exhibits network effects, as each new customer and application adds value for all of the others. Successful platforms also tend to tie the customer to the platform provider, because adopting the platform makes it easy for the customer to take advantage of the entire ecosystem of applications built on the platform. However, large-scale platforms tend to be developed later in the evolution of an industry. It can be quite challenging to scale up platforms before a critical mass of successful use cases and solutions (see next item) has developed.
- *The ability to provide end-to-end solutions*. Because of the degree of customization required to create effective IoT systems in specific industries and companies, companies that can supply complete solutions" — hardware, software, installation, and service — can establish deep relationships with customers that would be difficult for competitors to interrupt. End-to-end suppliers could be makers of equipment that uses IoT technology, software companies, or systems integrators.

The ability to provide complete solutions is one way that IoT suppliers could maintain or expand their position, even as the industry matures and their original advantages become less powerful. Indeed, the shift of value to software does not mean that companies that would traditionally be categorized as hardware suppliers will be at a permanent disadvantage—if they begin to craft solutions that include other parts of the value chain. Increasingly, equipment suppliers are incorporating IoT capabilities in their products and acting as software developers and consultants for

their customers, thereby providing end-to-end solutions. Similarly, software players are expanding their offerings to include hardware.

In the near term, there will be a premium for providing complex systems integration services, because of the custom nature of many IoT deployments. Systems integration skills are in high demand by IoT users and will likely remain so. Despite some efforts to promote interoperability, it is an extremely complex undertaking to bring together the devices and software that are needed in IoT systems. Moreover, customers need systems that are designed for their vertical markets (clothing store chains, for example) and customized for their specific needs. Today, creating industry-specific or custom IoT systems is labor-intensive and costly. By offering an integrated solution, suppliers can satisfy many customers' desire not to have to piece together solutions themselves (though some sophisticated buyers will). Over time, as the industry matures, we expect more packaged vertical solutions and large-scale horizontal platforms to take hold.

- **Where and how to compete: Horizontal and vertical expertise**

As companies develop IoT offerings, they will need to consider where to compete (what industries or settings to serve) and how to compete (which hardware, software, and services to offer). One issue that often arises is the degree to which their offerings should address a specific "vertical" niche vs. providing a broad "horizontal" platform. Creating complete vertical solutions requires integration across the value chain and a deep understanding of the industry and setting in which the solution will be deployed. This necessitates considerable investment, both in technology capability (even if done with partners) and in knowledge of the industry and setting. Meanwhile, providing horizontal platforms requires a critical mass of vertical solutions that satisfy specific customer needs in industries and settings. For other IT technologies, there seems to have been a time-dependent element to the evolution, as described above; a critical mass of vertical complete solutions is developed first, and then scaled platforms develop.

For IoT suppliers to grow as effectively as possible, one of their greatest challenges will be to understand which elements of a successful IoT implementation are applicable across different industries and to what degree each vertical and customer will require customization. For IoT suppliers attempting to refine their strategies, this comes down to understanding what is similar (and therefore translatable) among industries and specific customers and what differs.

As this point, we observe that while there are substantial opportunities for shared technology and capabilities across customer industries, the ability to offer general-purpose IoT software products is limited. Companies seeking to offer products across multiple industries will need to either specialize in very basic technology, such as hardware or connectivity, or develop industry expertise for each vertical they pursue as well as the capacity to customize vertical products for specific submarkets and customers. The opportunity for cross-vertical application of IoT technologies varies by the level of technology

While some elements of IoT software are easily shared—device management, data management, and security, for example—using the same algorithms to interpret data across different deployments is challenging. For example, even though many IoT applications need algorithms to perform common tasks, such as detecting patterns in streams of data, considerable tailoring is necessary to use the same algorithm in multiple contexts.

One widely used type of algorithm is for pattern recognition. This is basic to condition-based maintenance and health monitoring applications, both of which depend on determining the patterns in variables that are predictors of failure. The nuance driving these combinations can require a high degree of subject matter expertise to accelerate the process of creating predictive algorithms. Predicting the failure of an engine might require a combination of pressure, temperature, and engine vibration data. A system that can issue an alert only when readings on a mining truck are out of range might issue hundreds or thousands of alerts in a month, rather than identifying the handful that, when they appear in combination, indicate a need for preventive maintenance.

Even within an industry, a single algorithm that is not tailored to specific settings has limited application. ConocoPhillips spent five years developing an advanced software system to manage its complex of wells in the Ekofist field off the coast of Norway. However, translating that system to other locations even within ConocoPhillips would require substantial customization since some of the algorithms are specific to the geology of the Ekofist region. In some cases, transferring capabilities (the talent and experience of experts in IoT implementation, for example) is easier than transferring technology.

However, there are examples of algorithms that could be used in different IoT implementations, a few of which we describe here:

- *Pattern recognition of stream data*. Most of the routines used to find patterns in streams of data from IoT devices rely on well-known statistical techniques. The patterns that predict whether a truck engine or a gas turbine will break down are relatively similar. However, there are substantial differences that can require months of work by specialized data scientists and subject matter experts to tailor for specific applications. A new generation of algorithms being developed attempts to overcome this limitation using techniques from topology (the mathematical study of shape) and neural networks, especially deep learning.
- *Resource allocation*. Resource allocation problems (where to deploy employees, equipment, and other assets) are similar across settings, providing a way to maximize a specific outcome (plant output, for example) given a number of constraints (such as number of employees and time). The algorithms and math behind the optimization share a lot of similarities across settings, even when problems are large—some involve 20,000 constraints or more. The biggest difference between applications is calculating the impact of each constraint on the goal. Some impacts can be estimated using statistics and big data (how much a reagent increases yield in a pharmaceutical manufacturing plant, for example); others, however, will require knowledge of the business context and subject matter expertise (the additional value of assigning another employee to a construction site, for example).
- *2-D layout optimization*. Placement of goods and equipment in a 2-D space (a shop floor, for example) is a fairly common problem. The algorithms for each type of shop share a lot of similar characteristics. 2-D layout is a more specific version of the broader constrained optimization problem that resource-allocation algorithms attempt to solve.

- *3-D layout optimization*. In 3-D layout optimization, the challenge is to find the best way to pack a space, and the problem is far more complex. Maximizing the storage capacity of a truck or shipping container requires a very different set of algorithms than would be used to optimize placement of cereal boxes on a store shelf.
- *Path routing*. Path routing is used to determine how and where goods, equipment, and people should move most efficiently. This is one of the broadest algorithmic problems for IoT applications and is used in routing trucks, or employees in a warehouse. All share a number of similar elements. Given the large number of possible paths (to get from one part of New York to another that is 15 blocks up and 15 blocks over, there are 155 million possible paths), the common challenge is to select a smaller set of alternatives to make the computation efficient.
- *Computer vision and hearing*. Object recognition is the computer vision problem of distinguishing one object in an environment from another (identifying a table or a chair in a room). The underlying algorithms for computer vision are relatively similar, but a high degree of customization might be required if a system needs to be better at distinguishing a specific type of object. Distinguishing a dog from a cat, for example, can require much more customization than distinguishing a dog from a table. Sound analysis, for recognizing speech and other sounds, has similar requirements. Deep learning has also been successfully applied to these problems.
- *Emotion/mental state analysis*. Emotion analysis algorithms attempt to analyze the mental and emotional state of humans through computer vision and sound analysis. While the underlying algorithms can be built on top of those used for vision and speech, customization is required to adjust for context. The degree of customization varies. A safety system that needs to determine merely if a person is tired can be much more accurate with some customization than a general algorithm that needs to determine someone's broader emotional state, for example, to help a customer service representative to respond in an optimal way. The more emotional states a system needs to distinguish, the greater the nuance and customization required. Similarly, the more specific a distinction needs to be, the more customized the algorithm will need to be (distinguishing between excited and nervous for a casino is more complex than distinguishing between happy and sad for a retail store).